

MISSIONWELL AI — SIH DEFENSE Q&A; GUIDE

Predictive Stress & Welfare Monitoring System for Indian Armed Forces & CAPF

1. ELEVATOR PITCH & PROBLEM OVERVIEW

Q1: 60-Second Elevator Pitch

- **What It Is:** Defense-grade predictive welfare & operational readiness platform for Indian Armed Forces and CAPF.
- **Core Innovation:** Replaces reactive welfare with proactive early-warning intelligence using LightGBM AI & SHAP explainability.
- **Privacy Shield:** 100% DPDP Act 2023 compliant — zero ACR/APAR career prejudice.

Q2: What exact problem does this solve for MHA?

- **Current Gap:** 14-hr duty, extreme terrains (-35°C Siachen, LWE Bastar), and sleep debt. Existing checkups are reactive; Jawans suppress stress out of career fear.
- **Solution:** Delivers early risk indicators to Welfare Doctors while providing anonymized, aggregate readiness heatmaps to Commanders.

2. SYSTEM ARCHITECTURE & TECH STACK

Q3: Overall System Architecture

- **Frontend:** Next.js 16 (React 19, TypeScript, Tailwind CSS) with #090D16 military night-vision mode.
- **Backend API:** Node.js Express (TypeScript, Prisma ORM, Jose JWT auth, SHA-256 audit logging).
- **AI Microservice:** Python FastAPI (Port 8000) running LightGBM & SHAP TreeExplainer (<15ms latency).
- **Database:** SQLite (dev.db) for evaluation / PostgreSQL for production.

Q4: Why separate the Python ML microservice from Node.js?

- **Independent Scaling:** ML engine scales horizontally during surge assessments without impacting web traffic.
- **Ecosystem Standard:** Python provides native support for LightGBM, SHAP, and NumPy.
- **Low Latency:** FastAPI async execution returns predictions in under 15ms.

Q5: Multi-Branch Force Customization

- Dynamic adaptation of mottos, rank structures, badges, and terminology across **CRPF, Indian Army, BSF, ITBP, CISF, and State Police**.

3. MACHINE LEARNING & EXPLAINABLE AI (XAI)

Q6: Why LightGBM instead of Deep Learning or Logistic Regression?

- **Tabular Data Superiority:** Gradient Boosted Trees outperform neural networks on tabular physiological/duty telemetry.
- **Efficiency:** Handles complex non-linear feature interactions without GPU requirements.
- **Native XAI:** Direct support for exact SHAP TreeExplainer attribution.

Q7: Model Performance Metrics (77.9% Raw, 78.4% Balanced, F1 0.773)

- **Calibrated Precision:** Deliberately calibrated to model real-world physiological stress transitions.
- **No Overfitting:** 99% accuracy on clinical data indicates data leakage; 78% balanced accuracy guarantees reliable, un-inflated real-world risk detection.

Q8: SHAP Feature Attribution

- **Targeted Insights:** Breaks down risk scores into exact drivers (e.g. +32 pts Sleep Debt, +24 pts Continuous Deployment, +15 pts Leave Denial).
- **Actionable Care:** Enables doctors to prescribe targeted duty rotations instead of generic advice.

Q9: Anti-Symptom-Masking Guardrail

- **False-Bravado Detection:** Flags surveys completed in <15 seconds when baseline resting heart rate or duty hours are elevated.

4. DPDP ACT 2023 COMPLIANCE & PRIVACY

Q10: DPDP Act 2023 & Zero ACR/APAR Damage

- **Commander Isolation:** Individual survey responses and clinical notes return HTTP 403 Forbidden to Commander accounts.
- **Anonymized Command View:** Commanders see only battalion-level aggregate heatmaps.
- **Audit Ledger:** Every query is logged with SHA-256 hashes for System Admin auditing.

Role	Individual Data	Clinical Notes	Unit Heatmaps
Personnel (Jawan)	Own Data Only	Private Self-Care	Blocked (403)
Welfare Officer	Assigned Dossiers	Doctor-Patient Only	Full Triage View
Commander (CO)	Strictly Blocked	Strictly Blocked	Anonymized Unit
System Admin	Blocked	Blocked	Audit & System Health

5. UNIQUE DOMAIN FEATURES & TRICKY QUESTIONS

Q11: Buddy-Pair System (■■■■-■■■■)

- 1-tap confidential peer check-in ('■■■■ ■■ ■■■■■■■■ / ■■■■■■ ■■ ■■■■■■■■ ■■') to support buddies in high-stress zones.

Q12: Confidential Digital Sainik Darbar

- Direct digital request routing to bypass multi-tier rank hierarchy friction.

Q13: Remote / Zero-Connectivity Deployment (Siachen / Thar BOP)?

- **Answer:** Offline-first PWA architecture with local SQLite storage. Telemetry syncs in encrypted batches when network connectivity is restored.

Q14: What if a Commander demands individual mental health scores?

- **Answer:** Enforced HTTP 403 Forbidden at backend API level under DPDP Act Section 8(4). CO gets unit-level operational relief metrics only.

Q15: Does this replace military psychiatrists?

- **Answer:** No. It is a decision-support early-triaging tool that maximizes the efficiency and reach of limited medical personnel.

Q16: System Scalability across 10 Lakh CAPF Personnel?

- **Answer:** Stateless Node.js containers, PostgreSQL database indexing, and lightweight LightGBM microservice (~15ms response) ready for Kubernetes cluster deployment.

6. RECOMMENDED LIVE DEMO FLOW

1. **Google OAuth Login:** Select evaluator role (Doctor / Commander / Jawan).
2. **Jawan Assessment:** Complete assessment -> Trigger LightGBM score & SHAP drivers.
3. **Welfare Triage:** Doctor views confidential case & adds clinical notes.
4. **Commander View:** View anonymized battalion heatmap & download official MHA report.